

Tamweel — 90-Day Financing Default Risk Decision System

SDA-DSC-211 · Advanced Machine Learning Methods

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- Goal: estimate 90-day default risk using application-time information only.
- Synthetic teaching dataset: ~10,000 applications, 22 predictors, ~8% positive class.
- Decision quality is evaluated using ranking, cost, calibration and operational capacity—not accuracy alone.

1. Validation strategy

From baseline to honest evidence

- Removed post-application leakage fields before final validation.
- Used forward time splits, customer separation and 90-day label-maturity controls.
- Model comparison and threshold selection use out-of-fold predictions.
- Challenge data remains unlabeled and is not used for tuning.

Evidence	Day 1	Day 2	Day 3	Final
Main focus	Baseline	Honest validation	Cost/capacity	OOF selection
Key metric	AP / ROC-AUC	Forward OOF AP	Loss + capacity	Mean OOF AP

2. Final model selection

Keep the single Logistic Regression model

- Logistic Regression achieved the highest mean OOF Average Precision: 0.39166.
- No ensemble passed the worth-it improvement gate.
- Weighted ensemble was closest at 0.38942, still below the single model.
- The simpler model was retained because added complexity did not provide measured benefit.

Candidate	Logistic	XGBoost	LightGBM	Weighted
Mean AP	0.39166	0.35263	0.34549	0.38942
Decision	KEEP	No	No	No

3. Decision policy, calibration and explainability

Model quality must translate into an auditable policy

- OOF threshold: 0.16892; 11.37% of 2,155 OOF requests flagged, within the 12% capacity limit.
- Regional FPR diagnostic: Western 10.29% vs Eastern 6.31%; this is not a fairness certification.
- Calibration fit set: 836 requests / 78 positives; sigmoid did not improve fit-period Brier or ECE.
- SHAP and permutation importance are diagnostic explanations; SHAP is not interpreted causally.

4. Challenge delivery and limitations

Frozen model + frozen policy

- Challenge batch: 2,500 unlabeled requests.
- 330 requests exceeded the transported probability threshold; the 12% batch capacity retained 300.
- No AP, loss or fairness performance is claimed on challenge data because labels are unavailable.
- The project is educational only; synthetic data and simplified costs/capacity do not support real lending decisions.
- Monitoring plan: performance, calibration, capacity usage, score/data drift and regional diagnostics.